Velocity of the particle

$\vec{v} = \left(1.95\times10^{5}\,\hat{i} + 2.00\times10^{5}\,\hat{k}\right)\text{ms}^{-1}$

Charge of the proton $q = 1.60\times10^{-19}\text{C}$

Mass of the proton $m = 1.67\times10^{-27}\text{kg}$

(a) The force experienced by the proton is

$$\vec{F} = q(\vec{v}\times\vec{B})$$
$$= 1.60\times10^{-19}\times\left(\left(1.95\times10^{5}\,\hat{i} + 2.00\times10^{5}\,\hat{k}\right)\times\left(0.500\,\hat{i}\right)\right)$$
$$\vec{F} = 1.60\times10^{-14}\,\hat{j}\ \text{N}$$

Therefore, from Newton's second law,

$$\vec{a} = \frac{1}{m}\vec{F} = \frac{1}{1.67\times10^{-27}}\left(1.60\times10^{-14}\right)\hat{j}$$
$$= 9.58\times10^{12}\,\hat{j}\,\text{m s}^{-2}$$

(b) Trajectory is helical

Radius of helical path is

$$R = \frac{mv_z}{|q|B} = \frac{1.67\times10^{-27}\times2.00\times10^{5}}{1.60\times10^{-19}\times0.500}$$
$$= 4.175\times10^{-3}\,\text{m} = 4.18\,\text{mm}$$

Pitch of the helix is the distance travelled along x-axis in a time T, which is $P = v_x\,T$

But time,

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{|q|B} = \frac{2\times3.14\times1.67\times10^{-27}}{1.60\times10^{-19}\times0.500}$$
$$= 13.1\times10^{-8}\,\text{s}$$

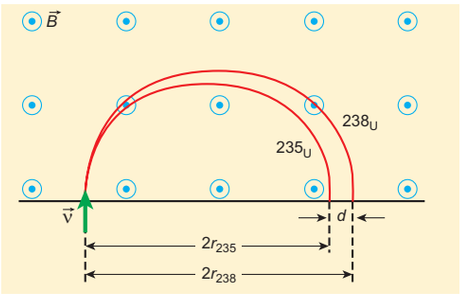
Hence, pitch of the helix is

$$P = v_x T = \left(1.95\times10^{5}\right)\left(13.1\times10^{-8}\right)$$
$$= 25.5\times10^{-3}\,\text{m} = 25.5\ \text{mm}$$

The proton experiences appreciable acceleration in the magnetic field, hence the pitch of the helix is almost six times greater than the radius of the helix.

EXAMPLE 3.21

Two singly ionized isotopes of uranium $^{235}_{92}U$ and $^{238}_{92}U$ (isotopes have same atomic number but different mass number) are sent with velocity 1.00×10^{5} m s^{-1} into a magnetic field of strength 0.500 T normally. Compute the distance between the two isotopes after they complete a semi-circle. Also compute the time taken by each isotope to complete one semi-circular path. (Given: masses of the isotopes: $m_{235} = 3.90\times10^{-25}$ kg and $m_{238} = 3.95\times10^{-25}$ kg)



Solution

Since isotopes are singly ionized, they have equal charge which is equal to the charge of an electron, $q = -1.6\times10^{-19}$ C. Mass of uranium $^{235}_{92}U$ and $^{238}_{92}U$ are 3.90×10^{-25} kg and 3.95×10^{-25} kg respectively. Magnetic field applied, $B = 0.500$ T. Velocity of the electron is 1.00×10^{5} ms^{-1}, then

(a) the radius of the path of $^{235}_{92}U$ is r_{235}

$$r_{235} = \frac{m_{235}v}{|q|B} = \frac{3.90\times10^{-25}\times1.00\times10^{5}}{1.6\times10^{-19}\times0.500}$$
$$= 48.8\times10^{-2}\,m$$
$$r_{235} = 48.8\ \text{cm}$$

The diameter of the semi-circle due to $^{235}_{92}U$ is $d_{235} = 2r_{235} = 97.6$ cm

The radius of the path of $^{238}_{92}U$ is r_{238} then

$$r_{238} = \frac{m_{238}v}{|q|B} = \frac{3.95\times10^{-25}\times1.00\times10^{5}}{1.6\times10^{-19}\times0.500}$$
$$= 49.4\times10^{-2}\,m$$
$$r_{238} = 49.4\ \text{cm}$$

The diameter of the semi-circle due to $^{238}_{92}U$ is $d_{238} = 2r_{238} = 98.8$ cm

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